Fiscal Year:	FY 2004	**Task Last Updated:**	FY 07/24/2006
PI Name:	Hassan, Yassin Ph.D.		
Project Title:	Multiphase Flow Studies Through Packed Beds Using Combined Particle Image Velocimetry and Laser Induced Flourescence		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77843-3133	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	08/01/2004	**End Date:**	07/31/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Motil, Brian J.	**Contact Phone:**	216-433-6617
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GB49G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Enhancing the understanding of fluid physics phenomena in the support of space flight is essential for basic life-sustaining functions and enabling technologies associated with space studies inside the spacecraft, or on planetary surfaces, to accomplish the human exploration of the space. One of the goals of this study is the charactization and understanding the phenomena of multiphase flow in packed beds. New advanced diagnostic techniques as particle image velocimetry and laser induced fluorescence will be utilized to obtain full field velocity and temperature of the bed. Predictive models and neural networks will be developed that aid in designing high performance of the beds.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	New project for FY2004.
Bibliography Type:	Description: (Last Updated:)